CardamomOT — analysis report

`/Users/eventre/Documents/GitHub/CardamomOT/experimental_datasets/Kameneva`

Generated on 2026-10-28 10:50

Run

Split	train
Stimulus penalisation	0.2
Prior penalisation	1.0
Genes	97
Stimuli	1
Observed cells	1260
Timepoints	0, 8, 32, 48, 64, 80
Cell types	precursors, intermediate, chromaffin, sympathoblasts
Network shown	inter_simul.npy
Embeddings	PHATE; mRNA: log1p=True, normalise=False, cell_depth_for_representation=True (counts / depth factor, model draws taken at the depth of the cells they mimic)

Perturbations (perturbation_simulation)

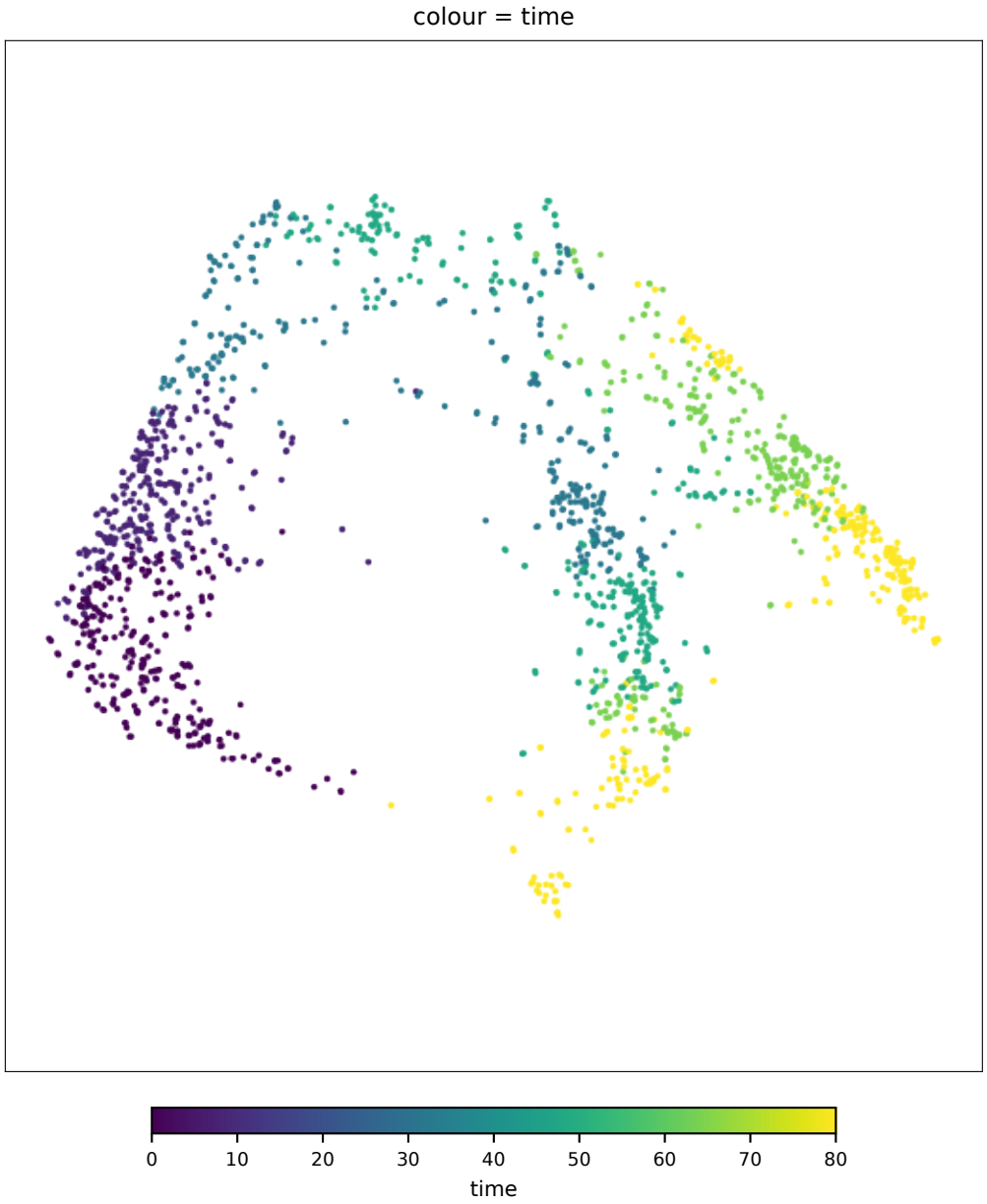
- ✓ P1 KO CHGA (50%)
- ✓ P2 KO CHGA
- ✓ P3 OV STMN2
- ✓ P4 KO CHGA · OV STMN2

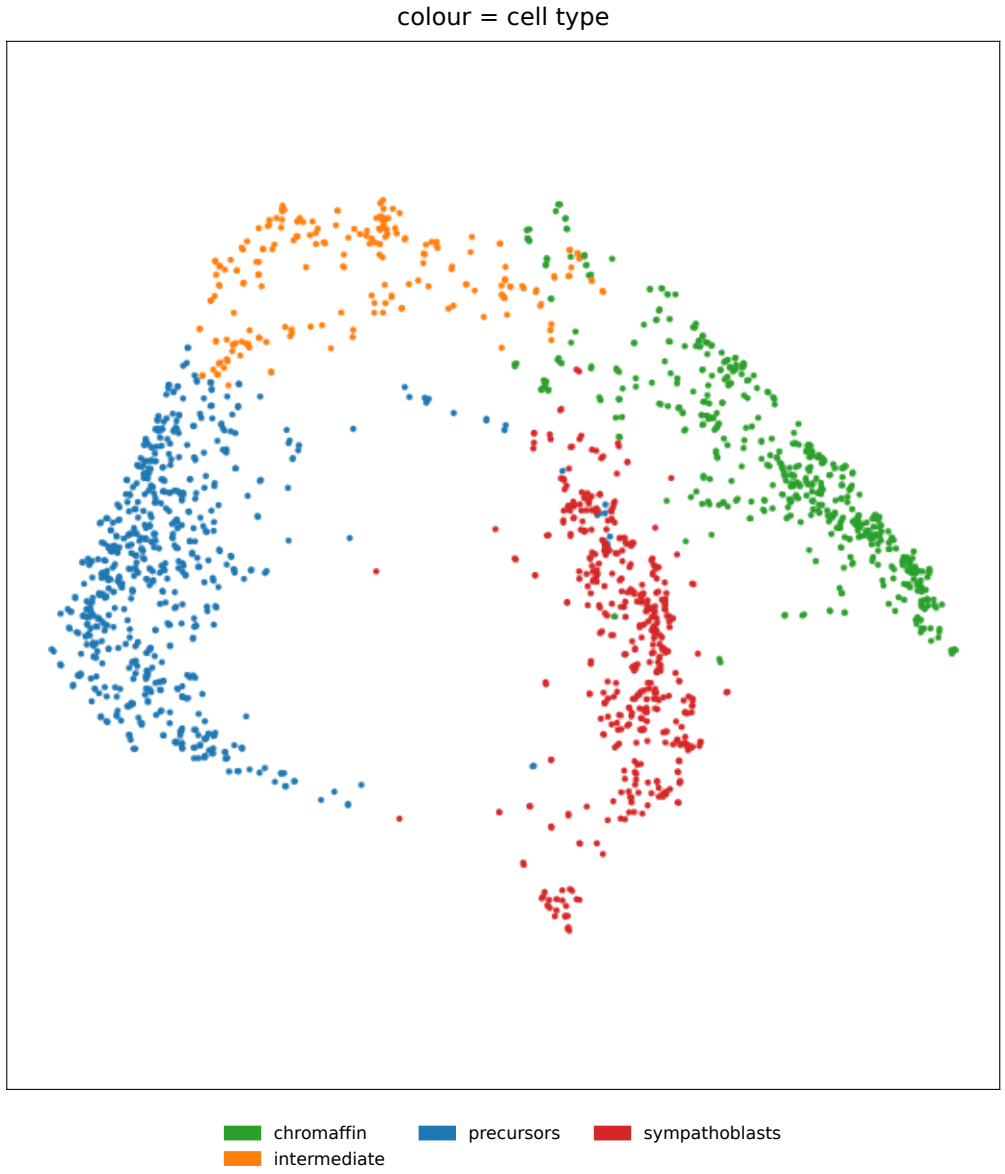
Contents

Data — UMAP of all the cells of Data/data.h5ad (time, samples), also on the genes of the model with select_genes; teaser: classical OT vs CardamomOT displacement fields (
1. Generative model — UMAPs of data, trajectories, NB mixture, network modes and simulation; cell-type proportions; gene-pair correlations; proteins
2. Gene regulatory network — regulatory power (violin plots), top-20 regulators and top-20 regulated genes (mutual top-10 edges)
3. In-silico perturbations — overview across KO/OV, then one page per perturbation
4. Proliferation — prior vs learned net rates, population growth, proteins driving growth; protein dilution at the birth rate
5. Learned dynamics — mRNA and protein fields (mechanistic velocity, displacement along trajectories), summary on mRNA
6. Held-out test cells — predictions with the network fixed vs the test data

Data — PHATE of all the cells

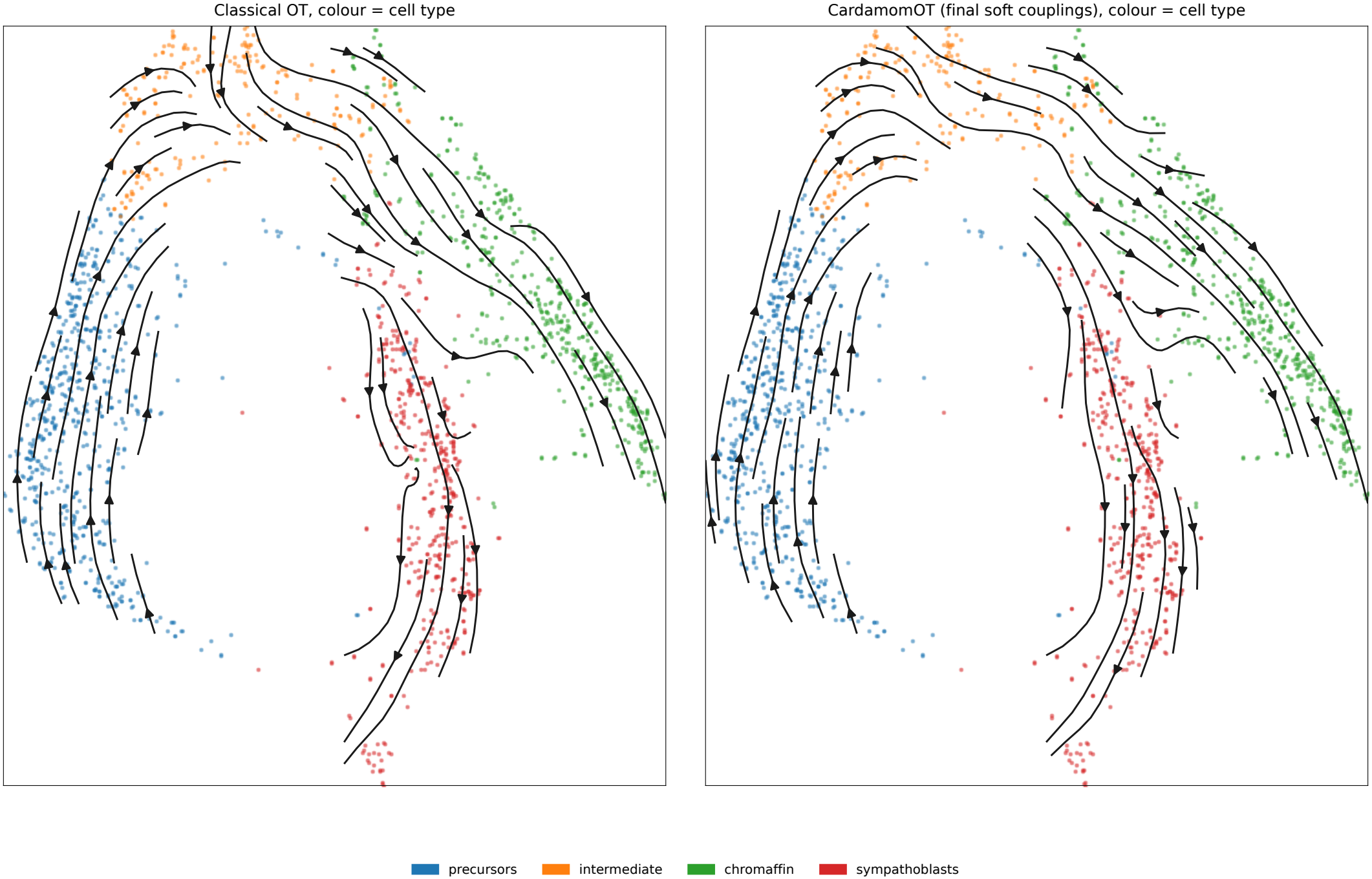
Data/data.h5ad, 1800 cells: log1p(counts), 97 highly variable genes, PCA (50 components), PHATE. Stored in cardamomOT/embedding_data.npz.





Teaser — displacement fields of classical OT and CardamomOT

Displacement to the mean descendant (x(t+1) − x(t), not divided by Δt), all times. Agreement on the cells with both direct displacements: weighted cosine 0.67 (PCA space). CardamomOT: final soft couplings of the trajectories (1260 cells); cells never reached or outside the training set: Gaussian kernel on the model mRNAs.



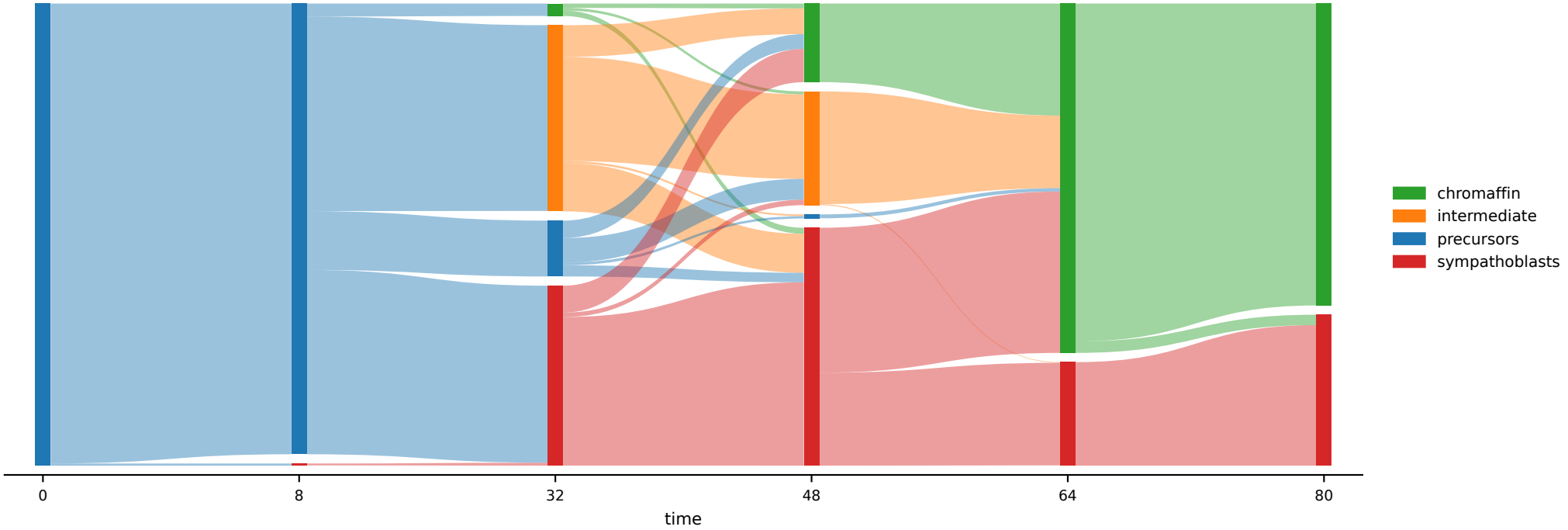
Teaser — displacement fields over time, classical OT vs CardamomOT

Cells of each t_from (coloured; other cells in grey) of the samples whose next timepoint is t_to, towards their mean descendant at t_to, kNN projection among their cells of t_from and t_to. cos: weighted cosine between the two methods on the cells of t_from with both direct displacements (PCA space).



Classical OT — cell-type transitions and fate genes

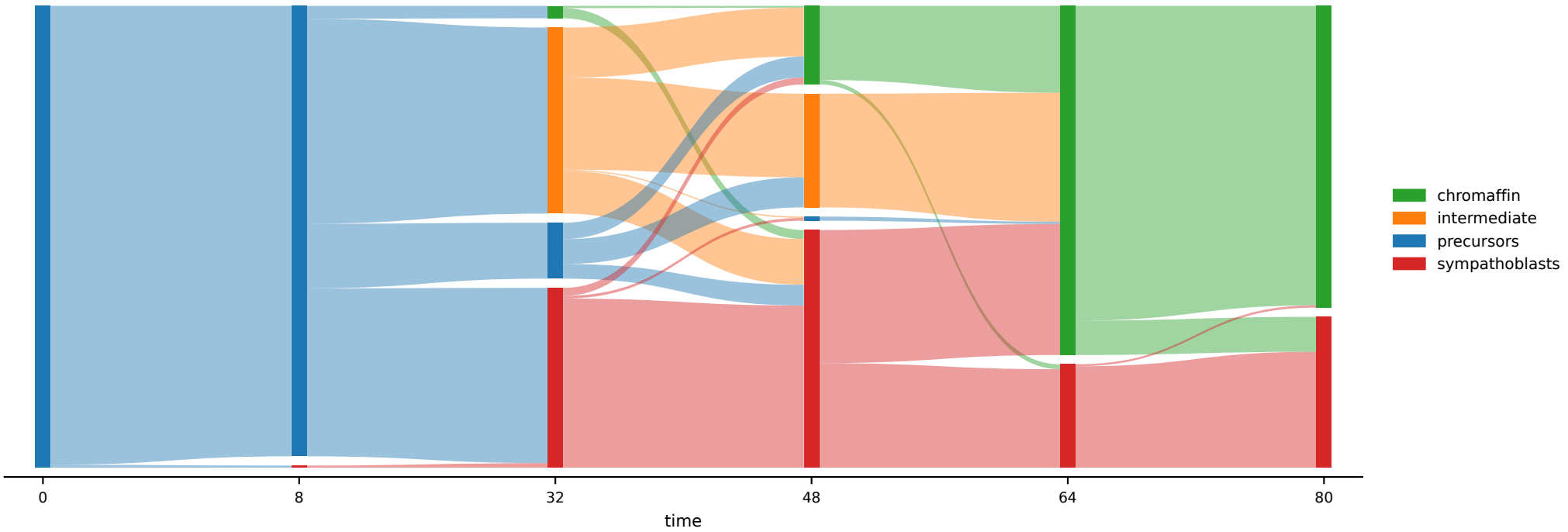
Waddington-OT couplings (every gene, train cells, per sample, growth from the net rate), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (classical_OT/fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
chromaffin	**CHGA** **CHGB** MEG3 **DLK1** **TH** **PCSK1N** **DBH** **RAMP1** **SCG2** RGS5	**CHGA** **DLK1** **CHGB** **RAMP1** **TH** **SCG2** HTATSF1 **PCSK1N** PENK **DBH**
intermediate	**PLP1** **POSTN** PTPRZ1 **COL5A2** **ZEB2** SPARC QKI EDNRB MPZ VIM	MT-CYB **POSTN** **ZEB2** COL1A1 VCAN **COL5A2** ALDH1A1 NRXN1 **PLP1** OLFML2A
precursors	**S100B** **VIM** **METRN** **PLP1** S100A10 **SPARC** ANXA2 **EDNRB** **MPZ** LGALS1	**PLP1** **S100B** **VIM** **EDNRB** **SPARC** PTPRZ1 **METRN** **MPZ** COL5A2 QKI
sympathoblasts	**PTMA** **GAP43** **RTN1** **STMN2** **BASP1** **TMSB10** **MAP1B** **UCHL1** **GAL** **PHOX2A**	**STMN2** **RTN1** **BASP1** **GAP43** **GAL** **PHOX2A** **MAP1B** **UCHL1** **TMSB10** **PTMA**

Teaser — CardamomOT: cell-type transitions and fate genes

Final soft couplings of the CardamomOT trajectories (model genes, data_<split> cells), pooled over the samples. Sankey: node = share of the cell type; links leave ∝ fate and arrive ∝ ancestor probabilities. Table: weighted Welch t, genes of both top 10 in red (cardamomOT/teaser_fate_genes.csv).



	Top-10 genes up in the ancestors (vs ancestors of the other cells)	Top-10 genes up in the cell type (vs the other cells)
chromaffin	**CHGA CHGB SCG2 TH DBH PCSK1N RAMP1** PENK **MEG3 DLK1**	**CHGA DLK1 CHGB TH MEG3 RAMP1 SCG2** HTATSF1 **DBH PCSK1N**
intermediate	**POSTN COL5A2** PTPRZ1 PLP1 **ZEB2 OLFML2A MPZ** EDNRB NRXN1 **VCAN**	**POSTN VCAN** NEFL **ZEB2** ALDH1A1 COL1A1 **OLFML2A COL5A2 MPZ** APOE
precursors	**S100B VIM PLP1** METRN ANXA2 **SPARC** S100A10 **EDNRB** ACTG1 IFITM3	**PLP1** PTPRZ1 **EDNRB SPARC** ZEB2 COL5A2 **VIM S100B** MPZ QKI
sympathoblasts	**GAP43 PTMA RTN1 STMN2 MAP1B BASP1 CD24 PHOX2A GAL** HAND2	**STMN2 RTN1 BASP1 GAL MAP1B GAP43 PHOX2A CD24 PTMA** UCHL1

Genes of the model (97)

Data/data_train.h5ad, alphabetical order.

ACTG1	EIF4A2	NEFL	RPL41
ALDH1A1	FAU	NPY	RPS13
ANXA2	FOS	NRXN1	RPS2
APOE	GAL	OLFML2A	RPS29
ATP5F1E	GAP43	PCSK1N	RPS4X
B2M	GAPDH	PENK	RTN1
BASP1	H2AFZ	PHOX2A	S100A10
BEX1	HAND2	PLP1	S100A6
BTF3	HAND2-AS1	PNMT	S100B
C4orf48	HINT1	POSTN	SCG2
CARTPT	HMGB2	PPIA	SPARC
CD24	HMGN2	PRPH	STMN2
CHGA	HNRNPA1	PTMA	TH
CHGB	HTATSF1	PTMS	TIMP3
CNTN1	IFITM3	PTN	TMSB10
COL18A1	IGFBP2	PTPRZ1	TPM1
COL1A1	ITM2B	QKI	UBC
COL1A2	LGALS1	RAMP1	UCHL1
COL5A2	MAP1B	RGS4	VCAN
CRYAB	MDK	RGS5	VIM
DBH	MEG3	RPL10	YBX1
DLK1	MEG8	RPL12	ZEB2
EDNRB	METRN	RPL30	
EEF1A2	MPZ	RPL32	
EEF2	MT-CYB	RPL34	

0. Per-cell sequencing depth

Diagnostic on the whole transcriptome within homogeneous groups (sample, time, cell type); counts are modelled as NB(k, c / s_i) (depth model 'mean') when the depth factor s_i is used.

Depth

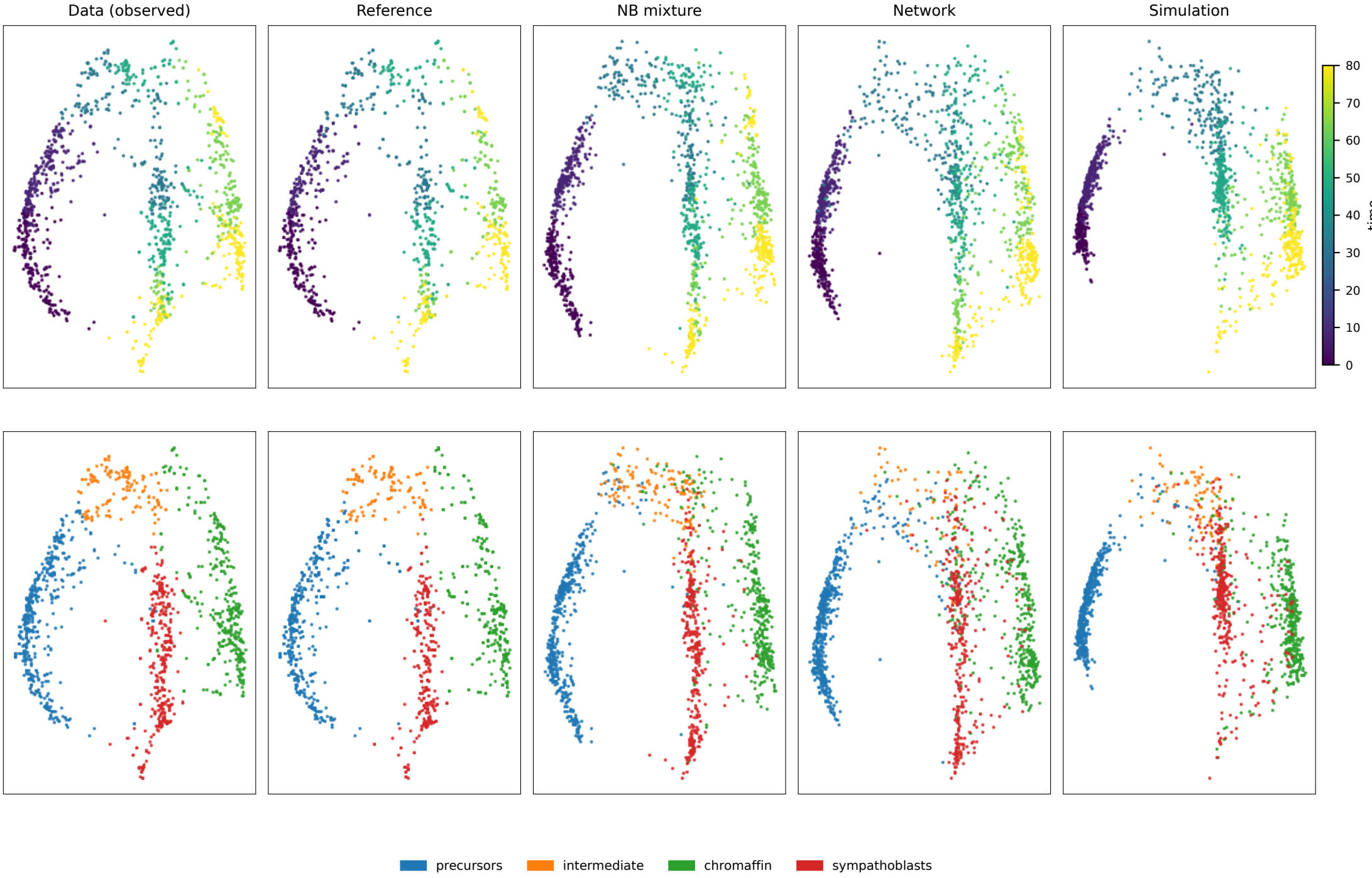
not estimated

Reason

97 genes < 10000

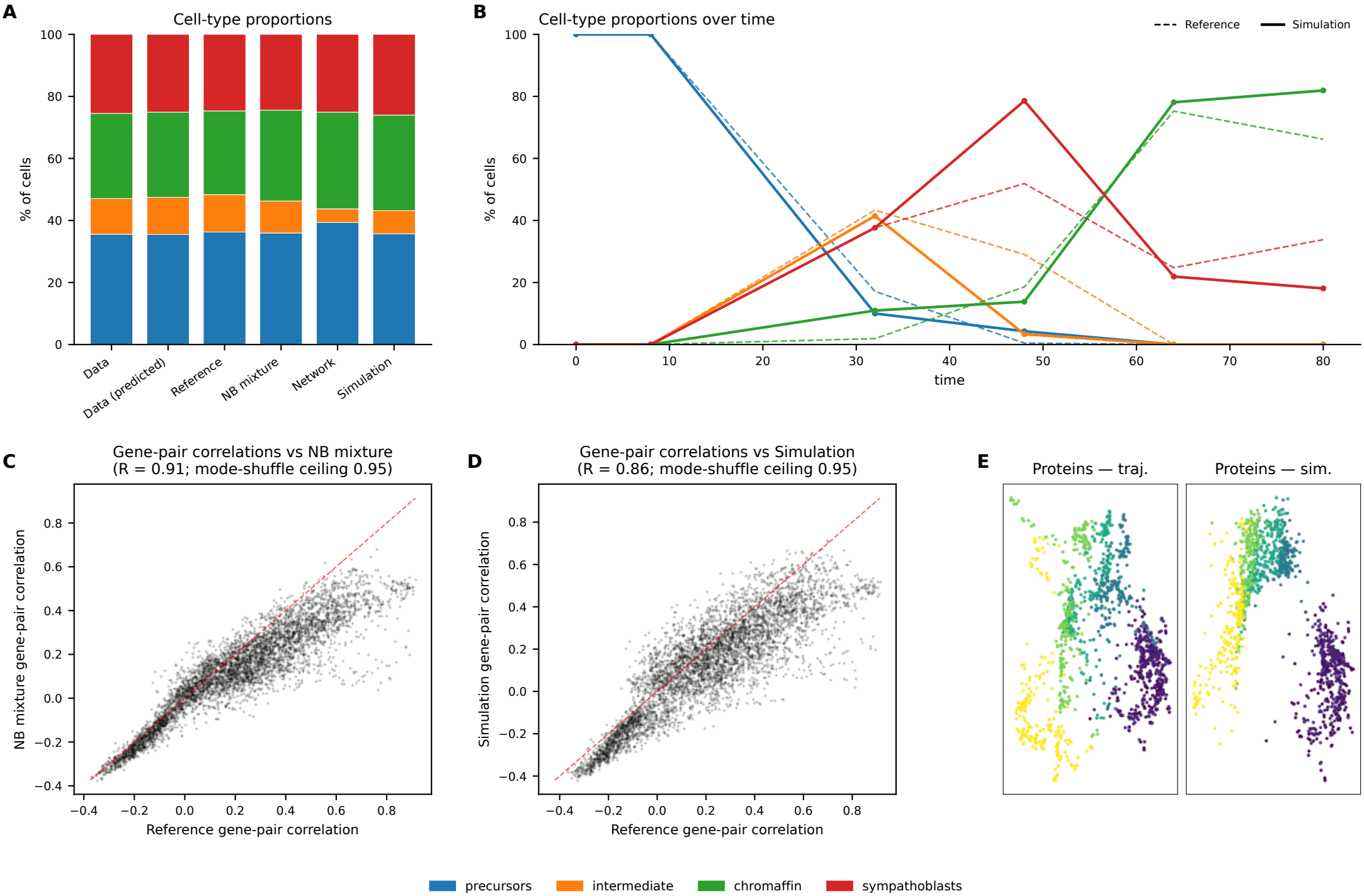
1. Generative model — trajectories and simulation

PHATE learned on the observed data; the trajectories (Reference: states drawn at each time among the observed cells), the NB mixture and network-driven modes along them and the full simulation (without proliferation) are projected onto it.



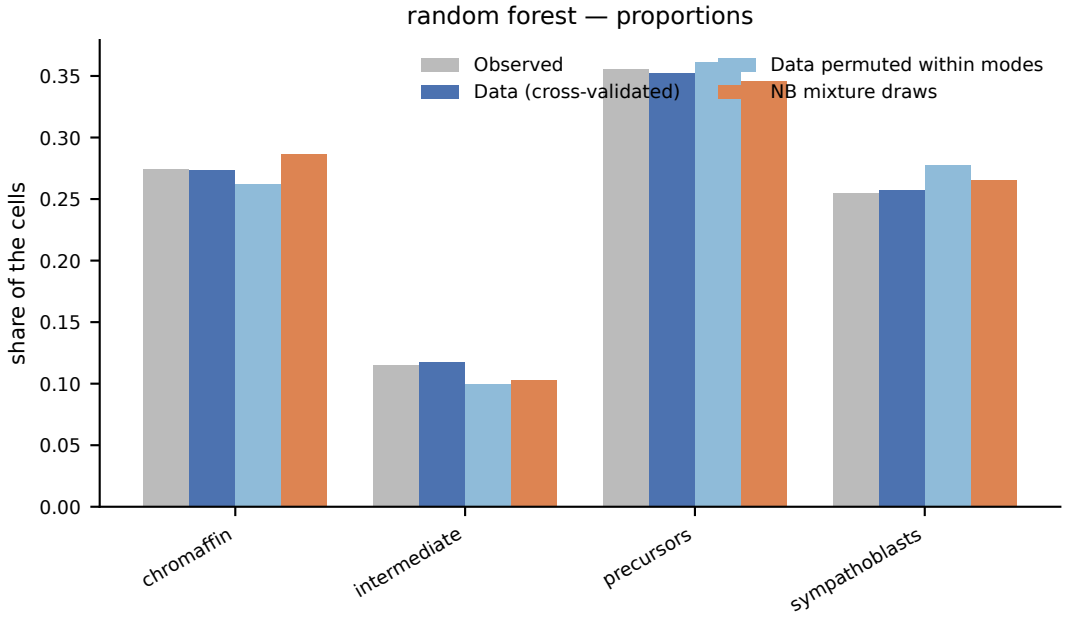
1. Generative model — quantitative checks

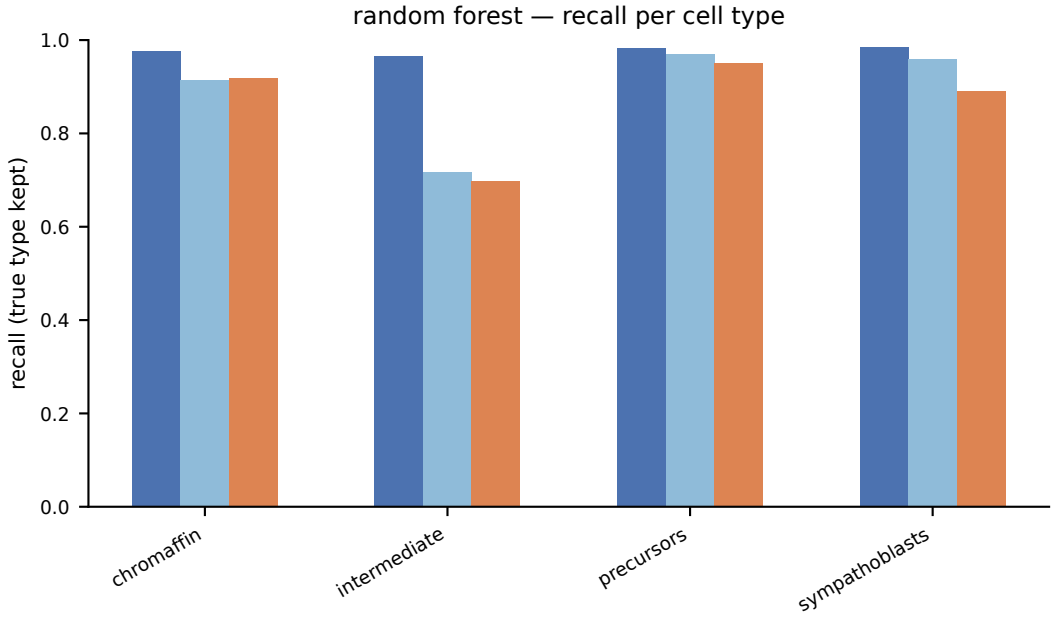
Reference = RNA of the trajectories (states drawn at each time among the observed cells: the observed composition); simulation without proliferation.

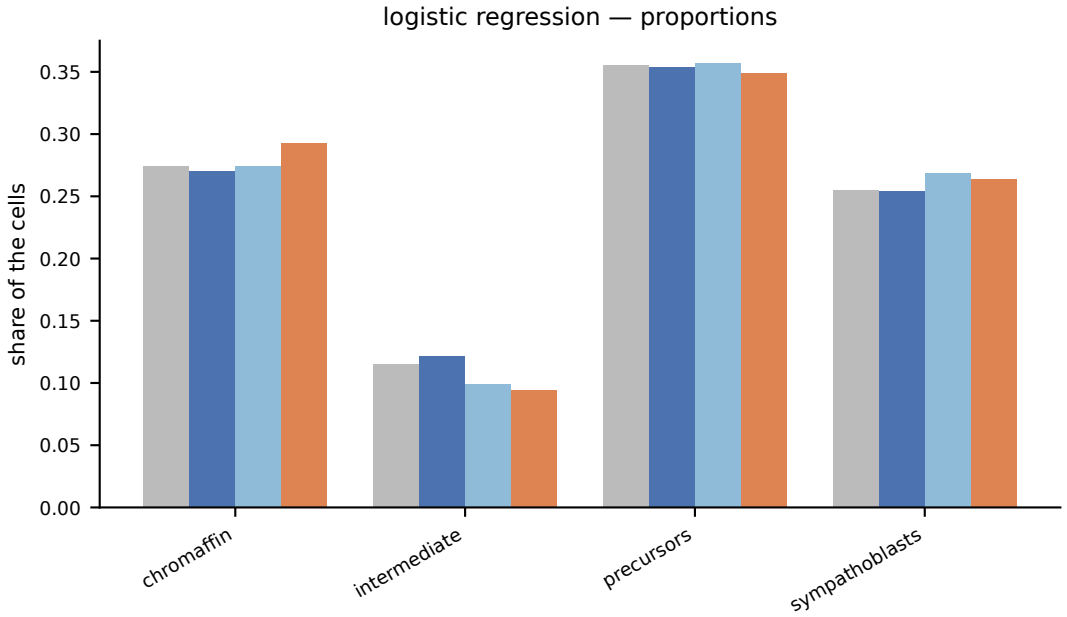


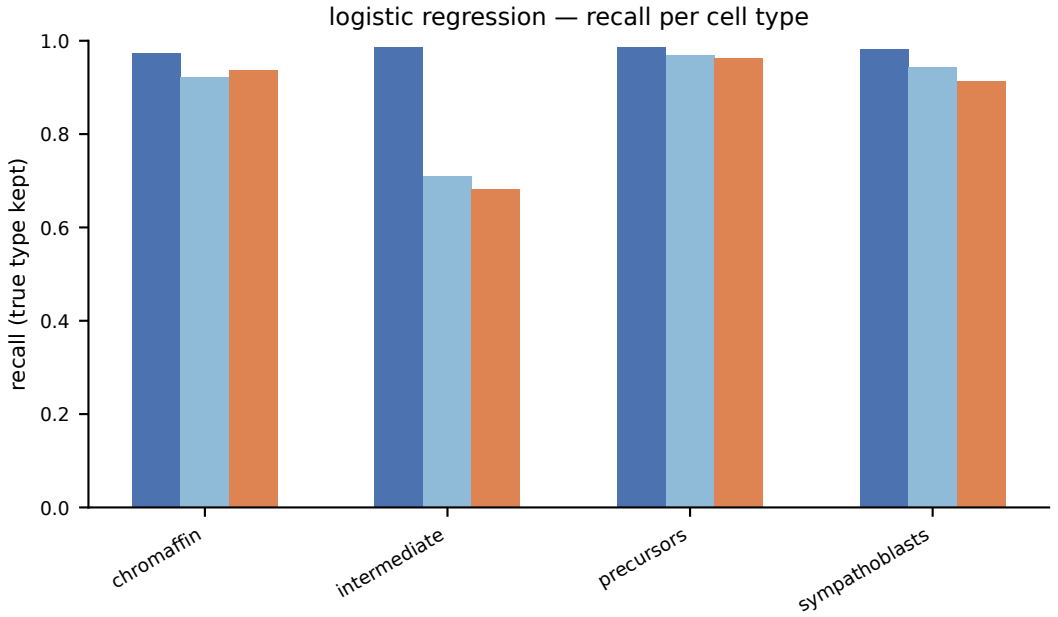
1. Generative model — cell-type identity in the NB mixture

One draw per real cell, per-sample classifiers trained on the other cells. Cross-validated data = classifier limit; permuted within modes (same sample, time, mode) = best model with these modes. Gap data → permuted: modes too coarse; permuted → draws: NB fit.





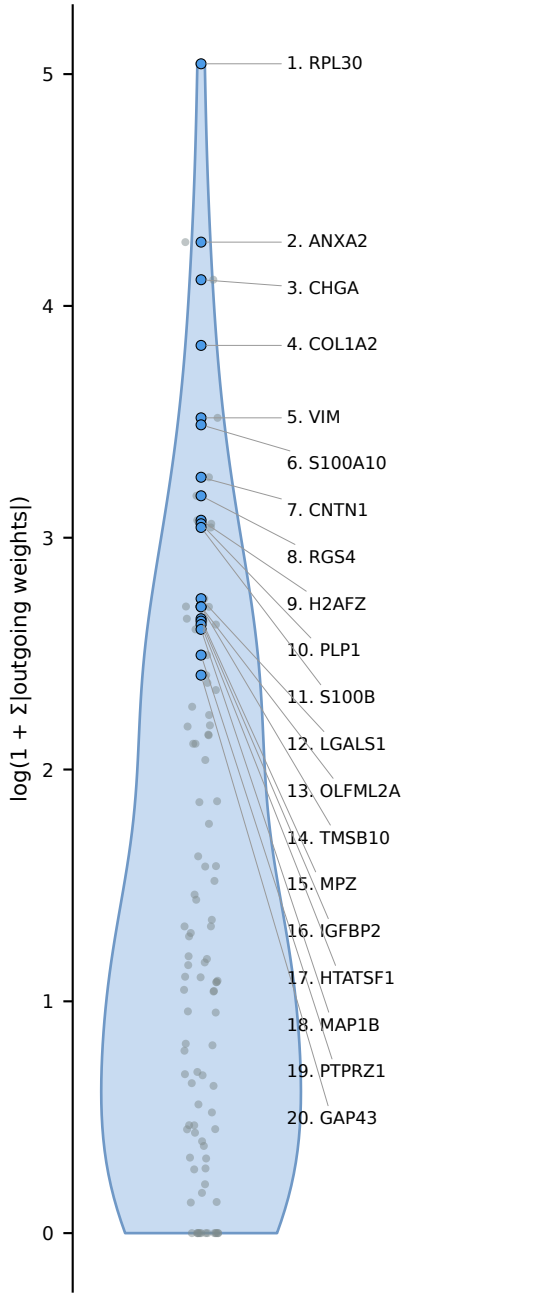




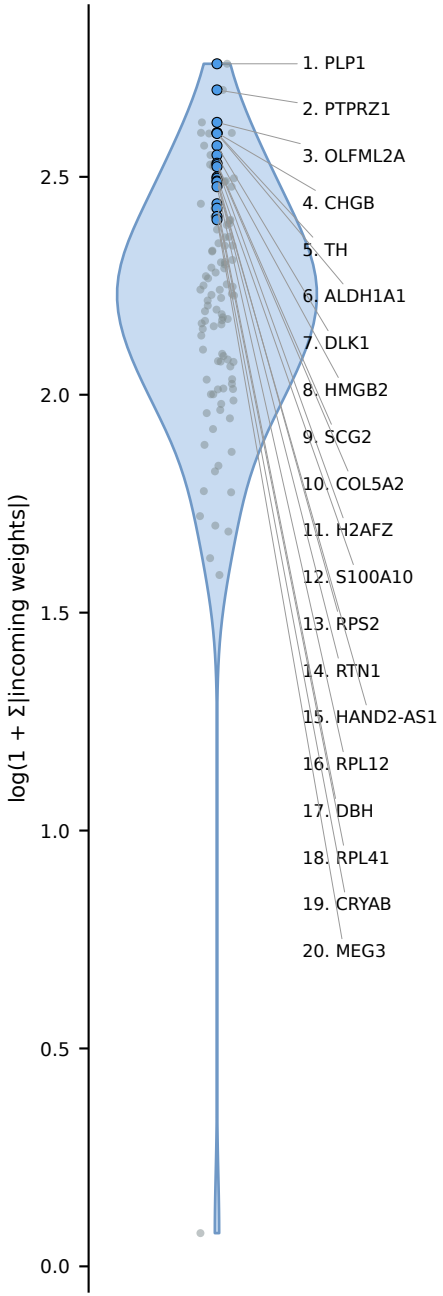
2. Gene regulatory network — regulatory power

Self-interactions excluded. Top 20 genes labelled; stimulus not shown (stimulus = 0.2 < 0.5).

A



B



C Top 20 regulators

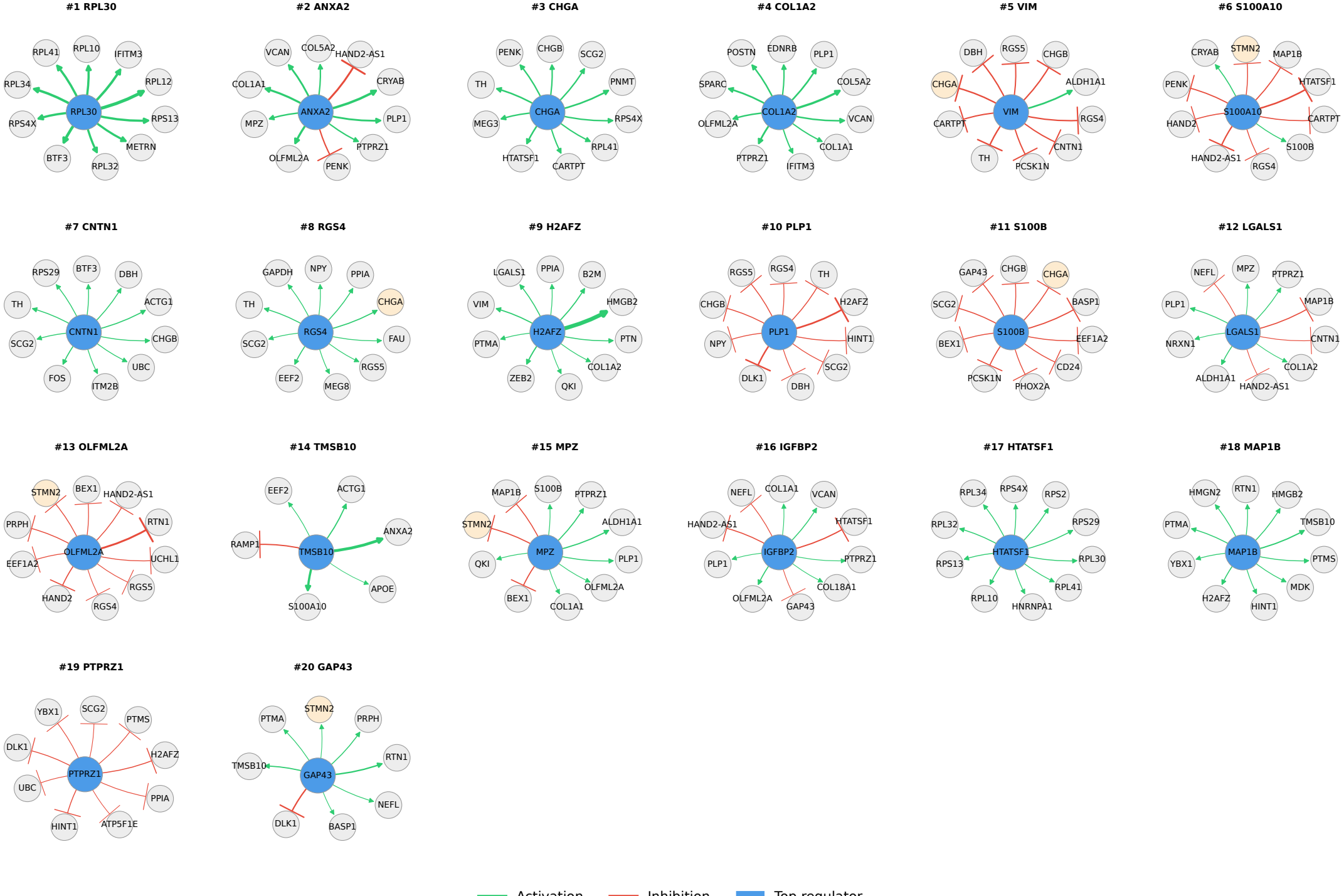
#	Regulator	Σ|w|	act.	inh.
1	RPL30	154.18	63	3
2	ANXA2	70.89	25	23
3	CHGA	60.13	40	2
4	COL1A2	45.05	20	8
5	VIM	32.69	3	21
6	S100A10	31.69	21	27
7	CNTN1	25.07	39	0
8	RGS4	23.07	36	0
9	H2AFZ	20.66	14	0
10	PLP1	20.33	2	21
11	S100B	19.99	1	22
12	LGALS1	14.44	17	12
13	OLFML2A	13.93	1	15
14	TMSB10	13.91	5	1
15	MPZ	13.16	12	6
16	IGFBP2	13.02	18	6
17	HTATSF1	12.81	15	0
18	MAP1B	12.52	15	0
19	PTPRZ1	11.10	0	21
20	GAP43	10.10	8	1

D Top 20 regulated genes

#	Target	Σ|w|	act.	inh.
1	PLP1	14.79	11	0
2	PTPRZ1	13.87	12	0
3	OLFML2A	12.80	8	0
4	CHGB	12.48	5	7
5	TH	12.47	5	8
6	ALDH1A1	12.46	10	0
7	DLK1	12.09	4	8
8	HMGB2	11.81	4	1
9	SCG2	11.57	4	6
10	COL5A2	11.53	12	0
11	H2AFZ	11.47	8	5
12	S100A10	11.18	7	0
13	RPS2	11.14	12	0
14	RTN1	11.07	5	2
15	HAND2-AS1	11.04	0	9
16	RPL12	10.91	7	0
17	DBH	10.45	4	6
18	RPL41	10.34	10	0
19	CRYAB	10.12	7	0
20	MEG3	10.04	7	7

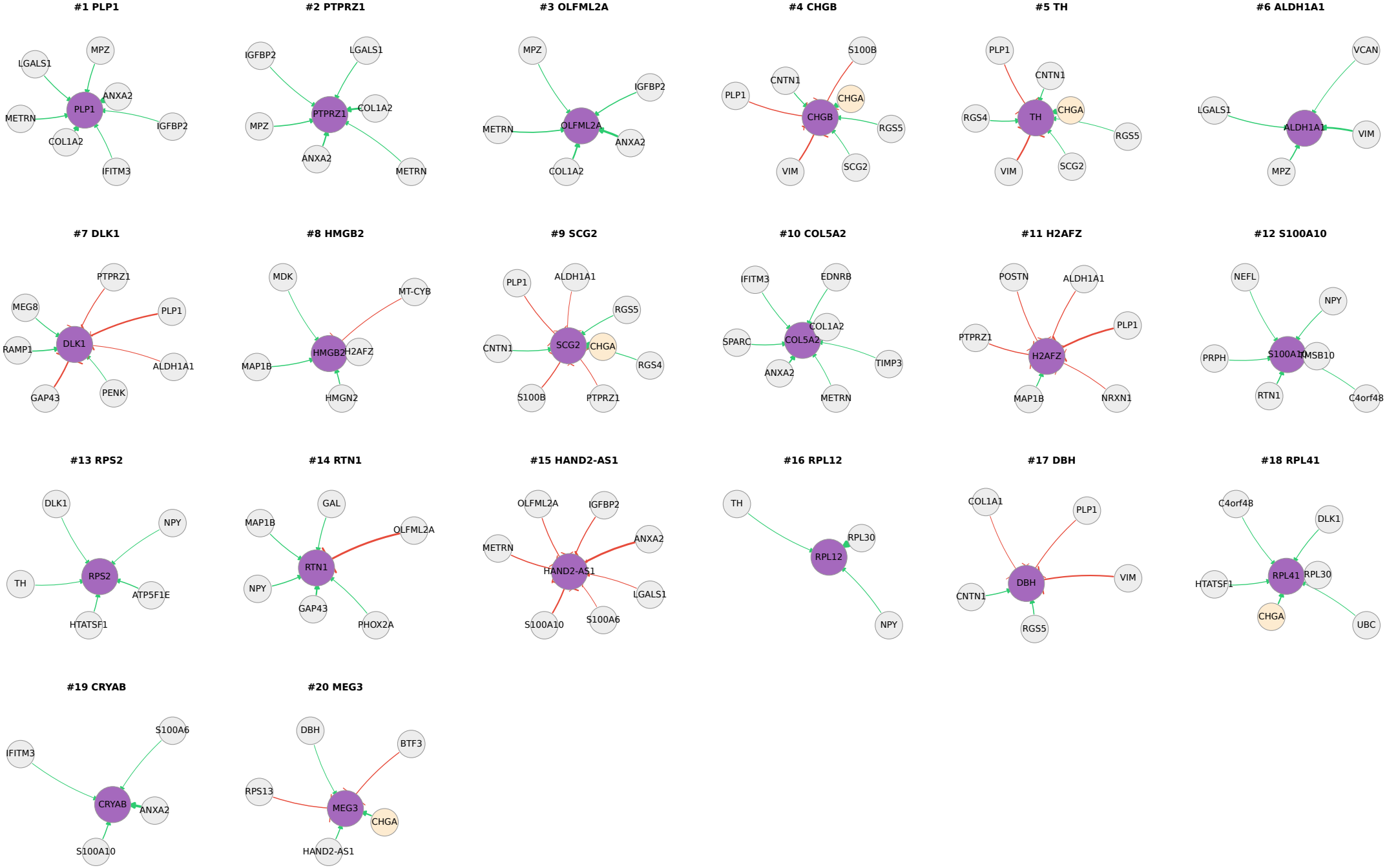
2. Gene regulatory network — top regulators and their main targets

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



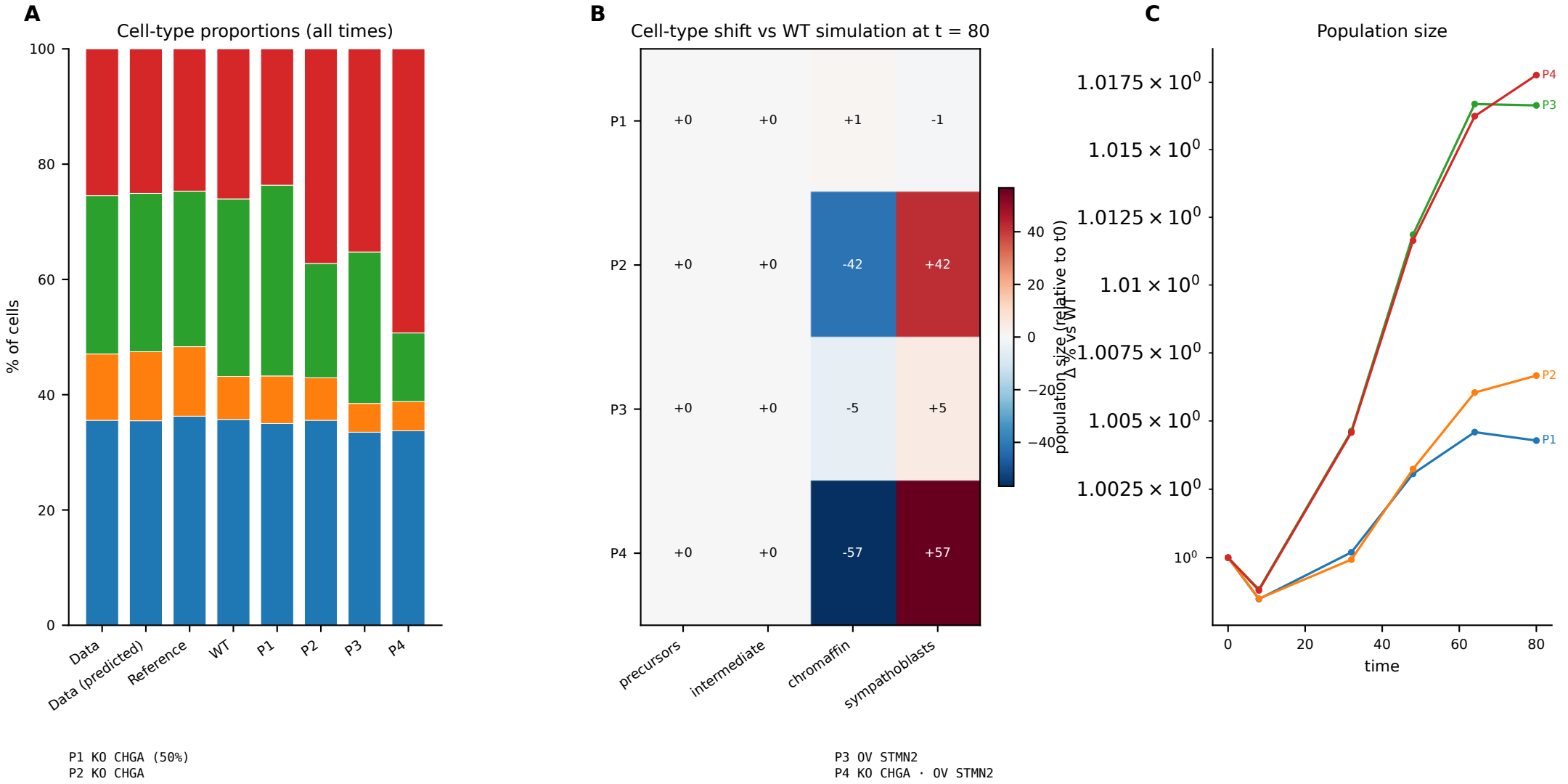
2. Gene regulatory network — top regulated genes and their main regulators

Edges shown only if among the 10 strongest (|w|) targets of the regulator AND the 10 strongest regulators of the target; edge width ∝ |weight| (global scale).
Perturbed genes (KO/OV) are highlighted in beige.



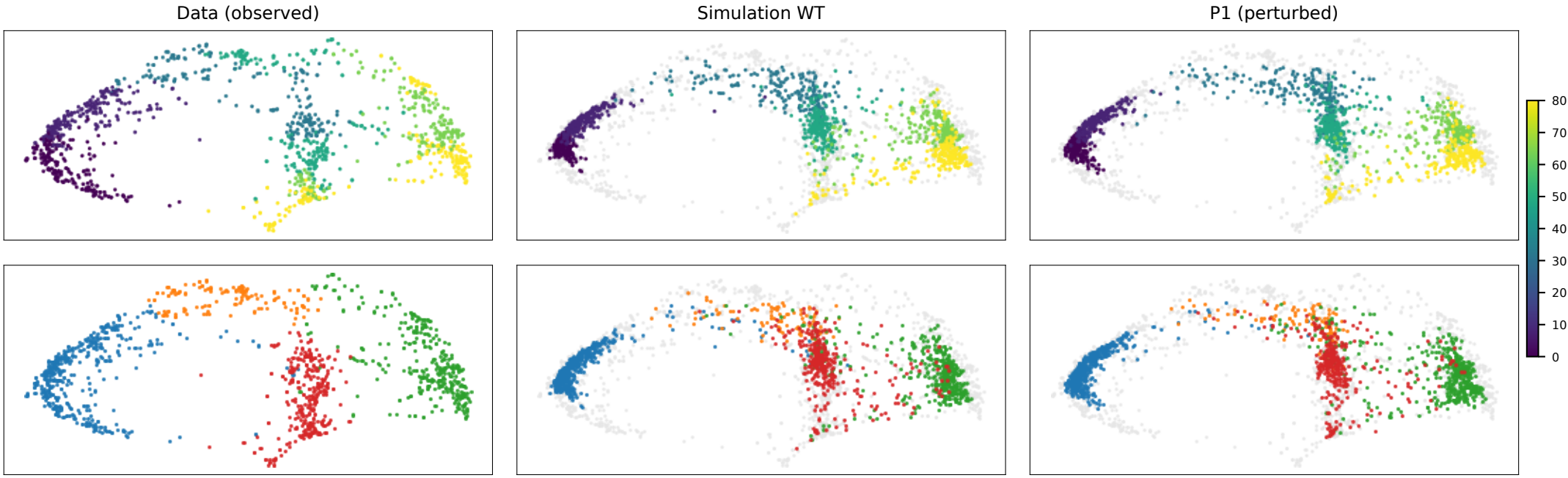
3. In-silico perturbations — overview

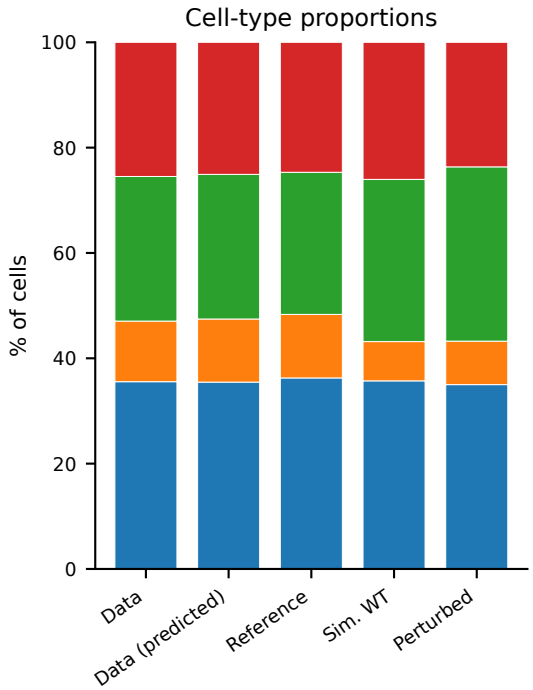
Cell types predicted by per-sample classifiers (logistic) trained on the observed data. Population sizes from the branching simulation (net rates + RATE effects).

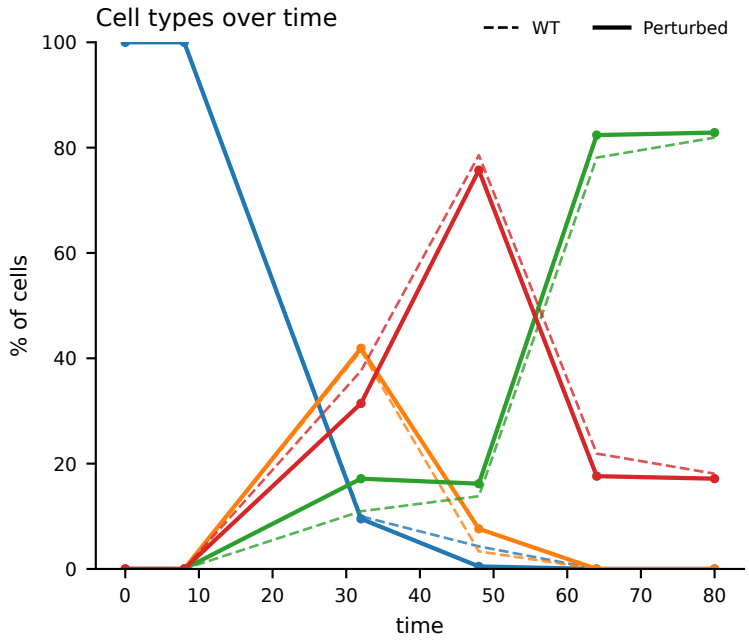


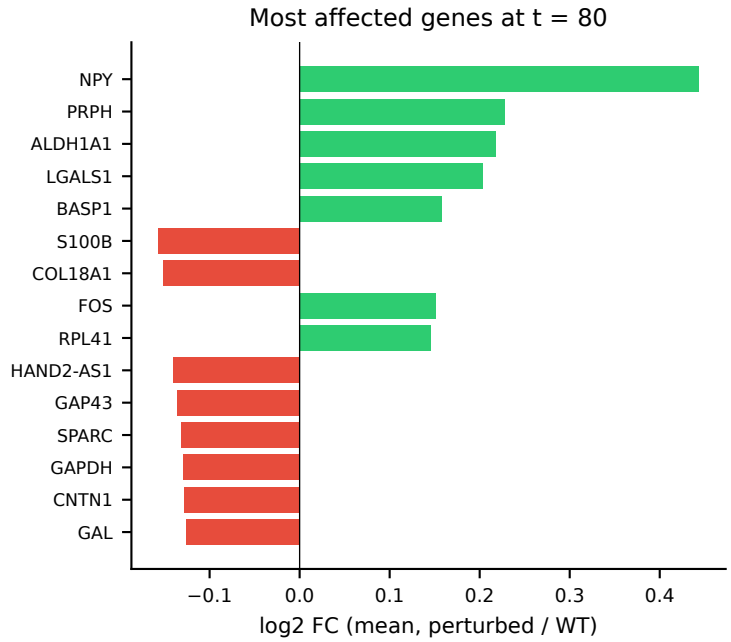
3. Perturbation P1 — KO CHGA (50%)

KO_CHGApct50_OV_none. PHATE: perturbed simulation projected on the WT embedding (grey = observed data).





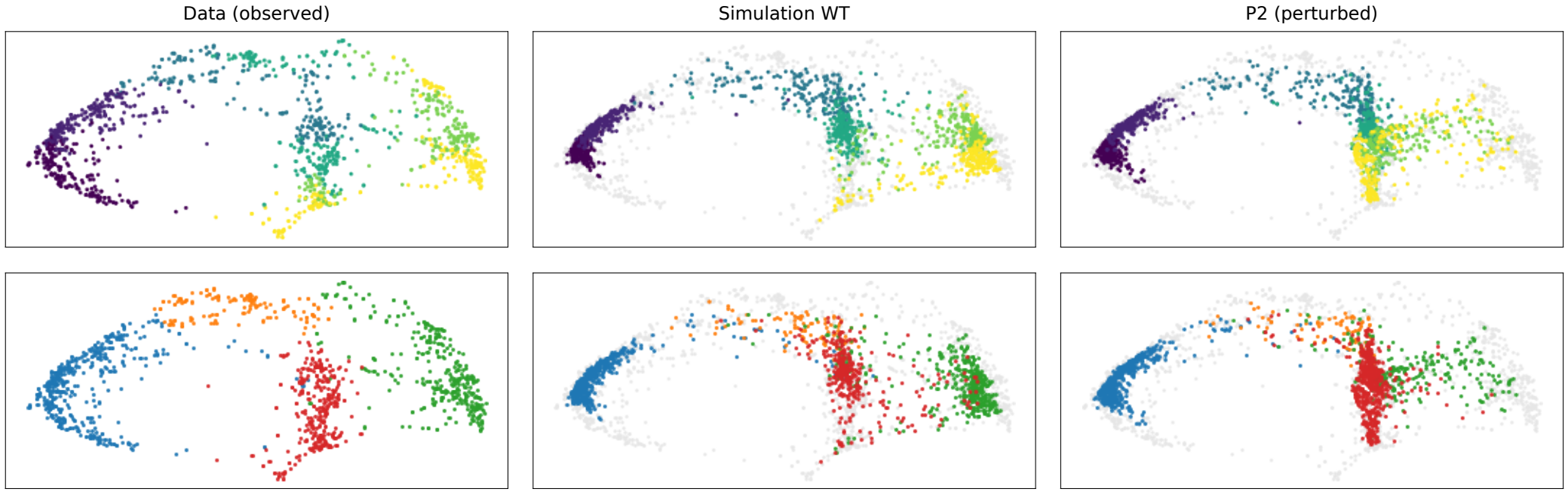


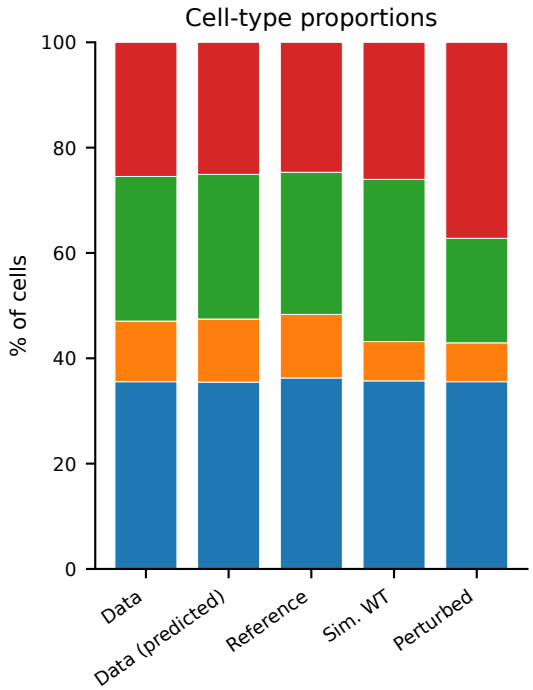


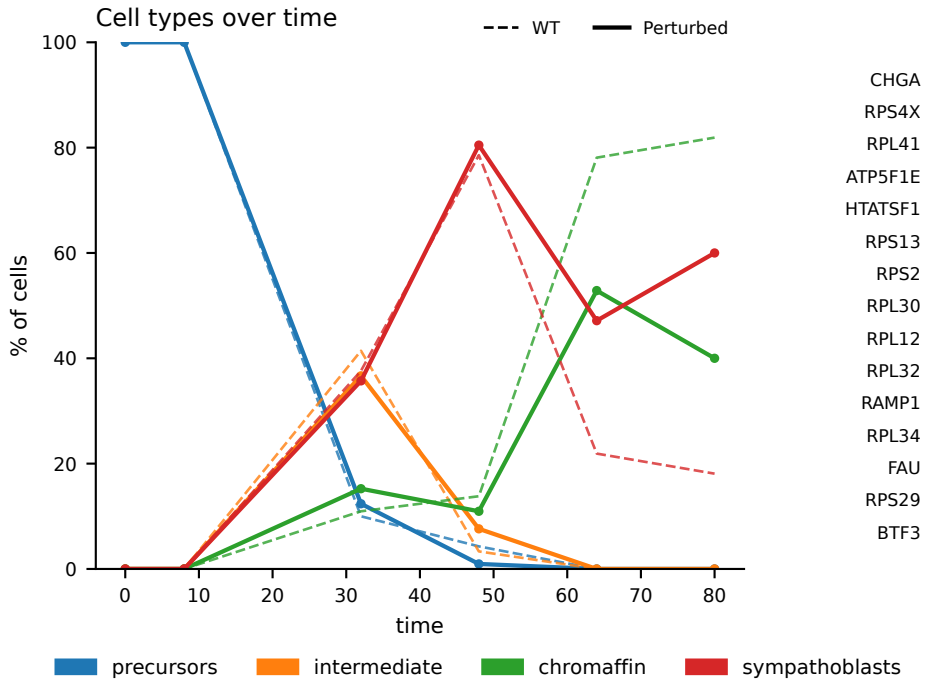


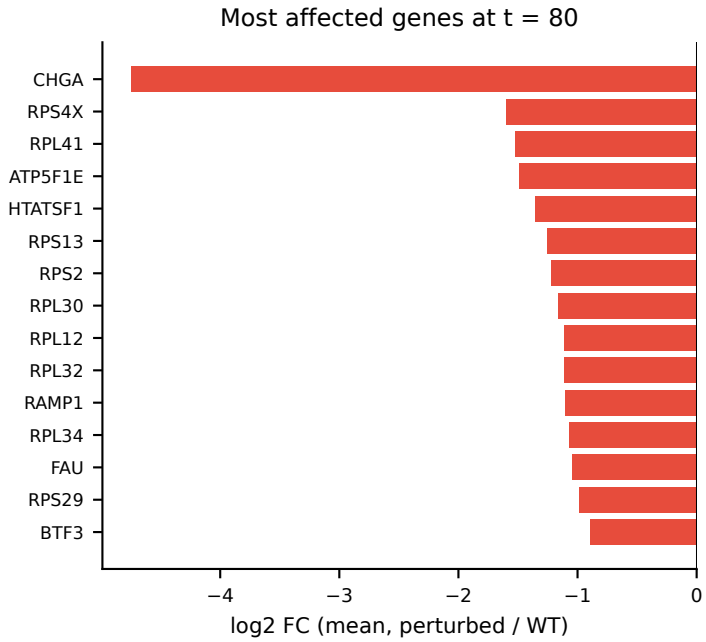
3. Perturbation P2 — KO CHGA

KO_CHGA_OV_none. PHATE: perturbed simulation projected on the WT embedding (grey = observed data).



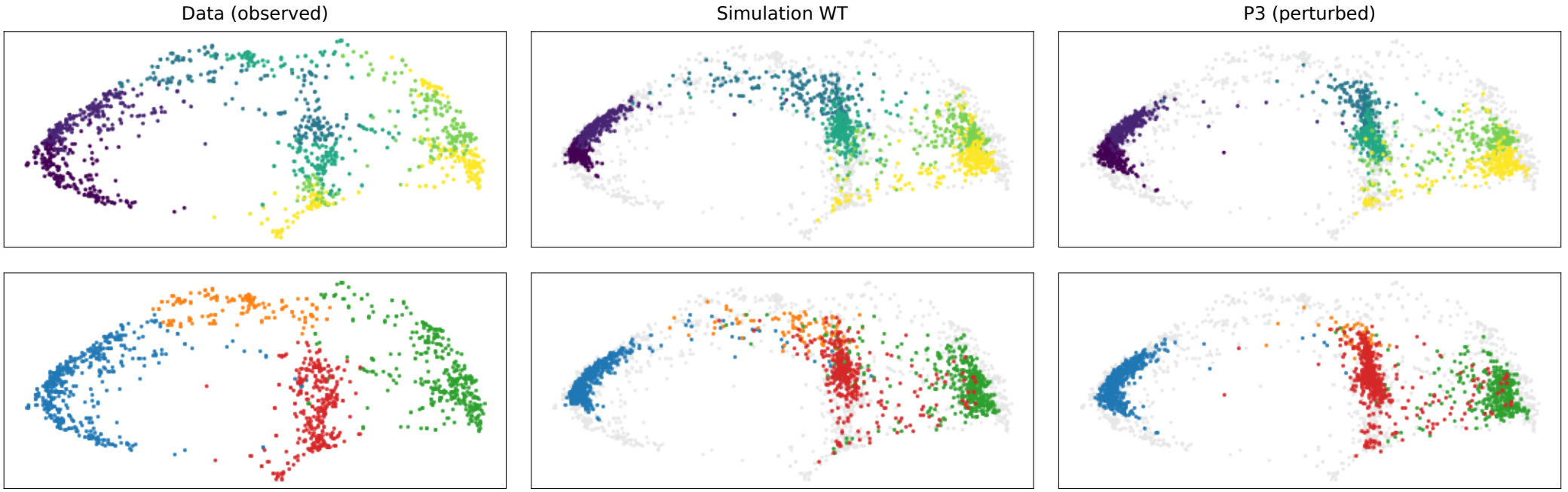


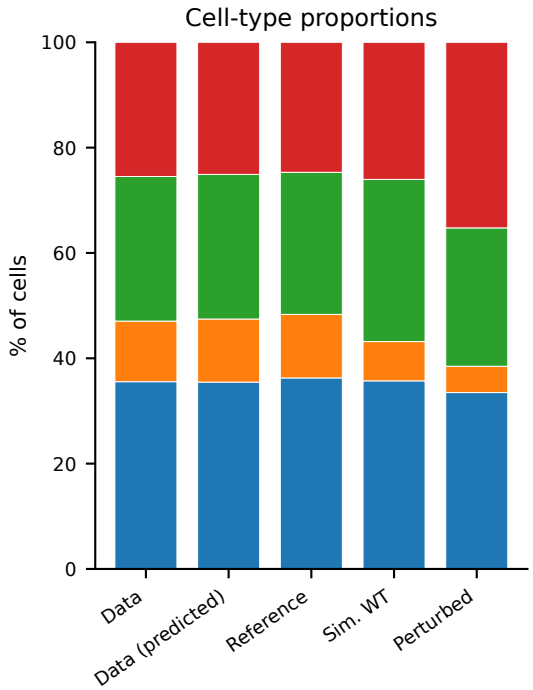


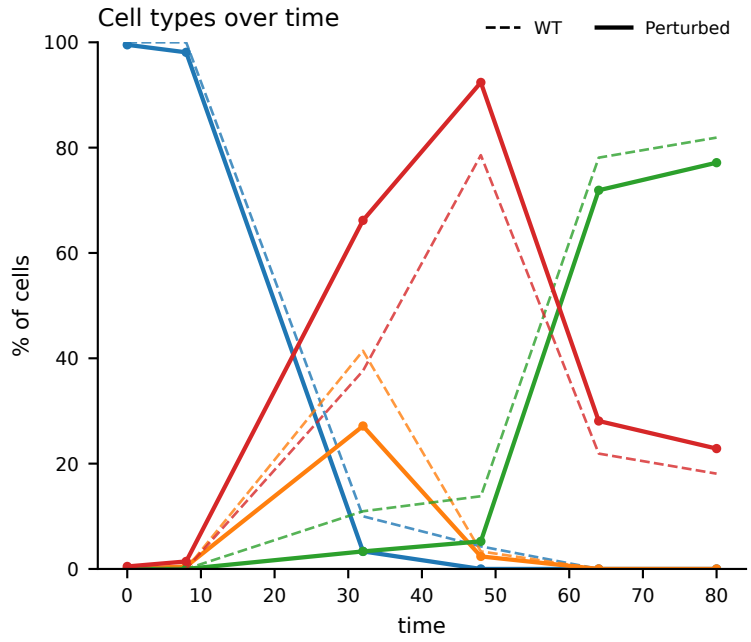


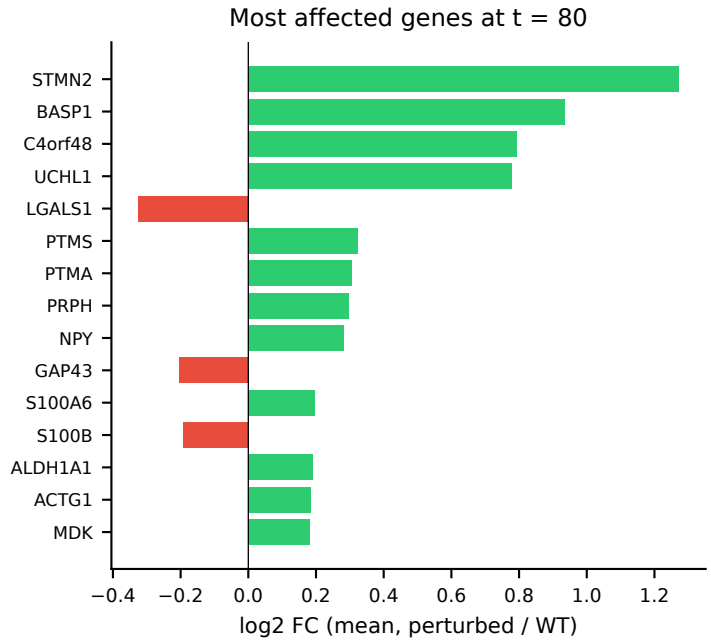
3. Perturbation P3 — OV STMN2

KO_none_OV_STMN2. PHATE: perturbed simulation projected on the WT embedding (grey = observed data).





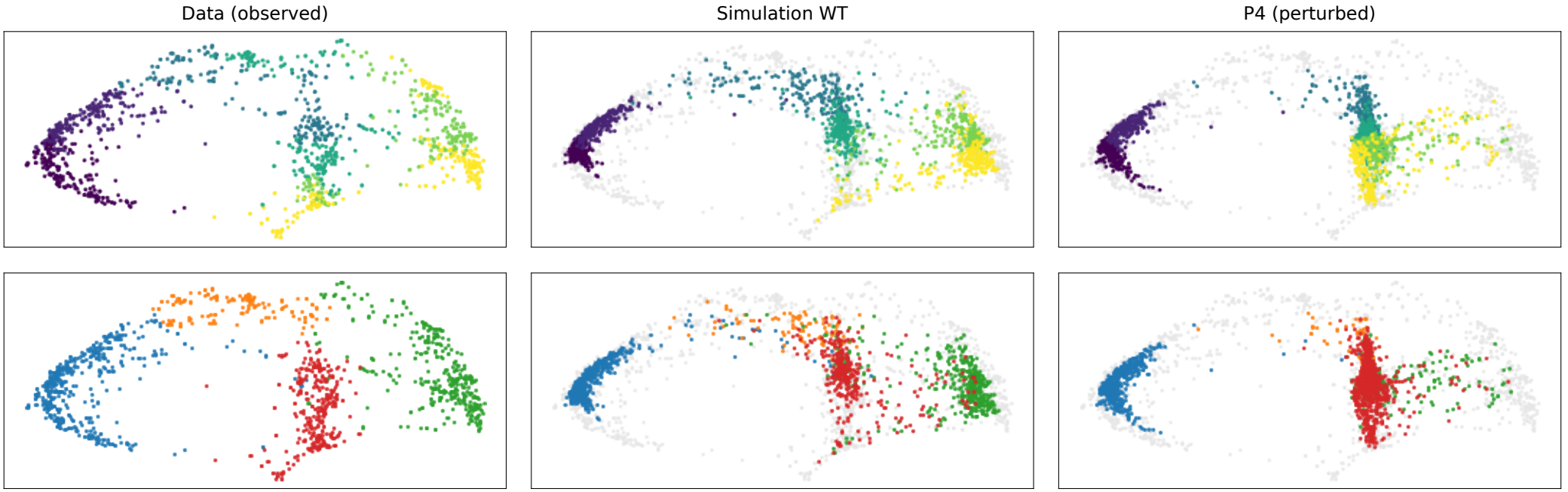


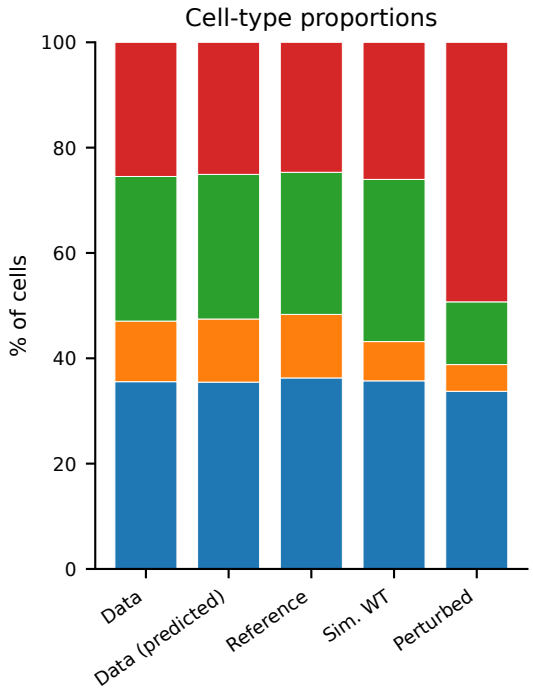


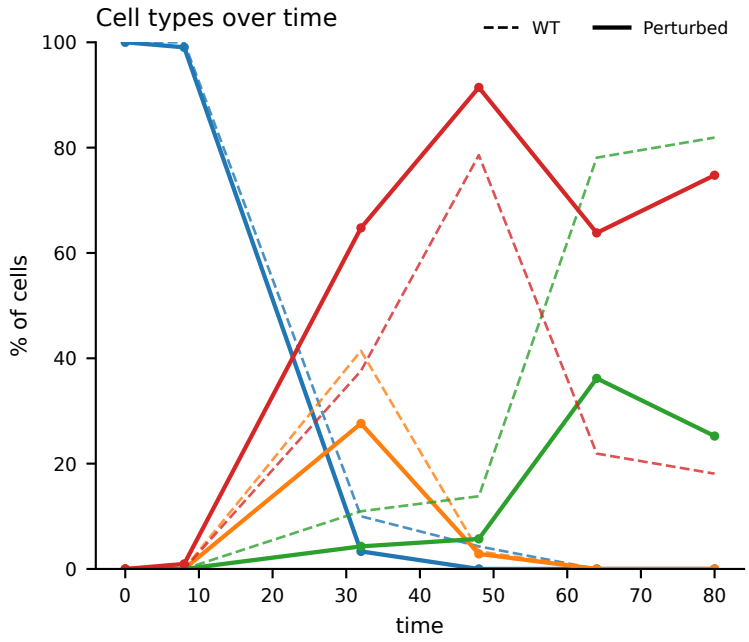


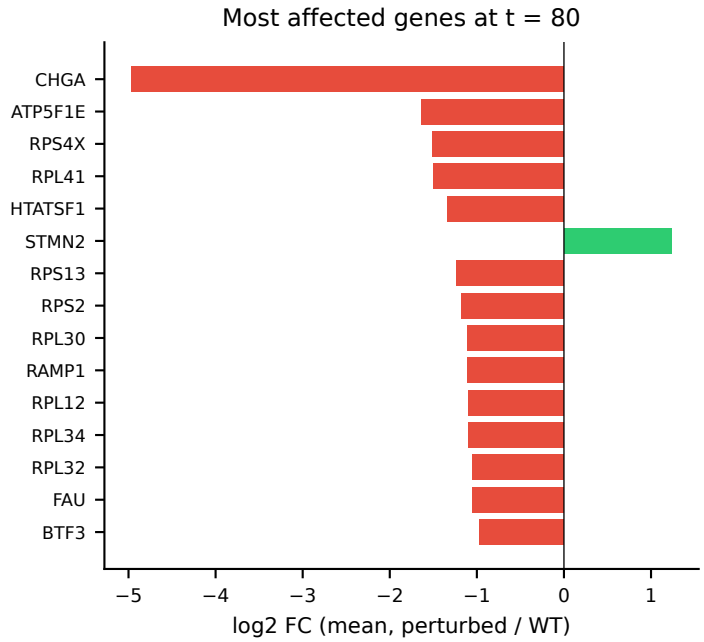
3. Perturbation P4 — KO CHGA · OV STMN2

KO_CHGA_OV_STMN2. PHATE: perturbed simulation projected on the WT embedding (grey = observed data).











4. Proliferation

Section unavailable:

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FileNotFoundError('no growth estimate (data_R_opt.npy / data_R_prior.npy from infer_network_structure)')
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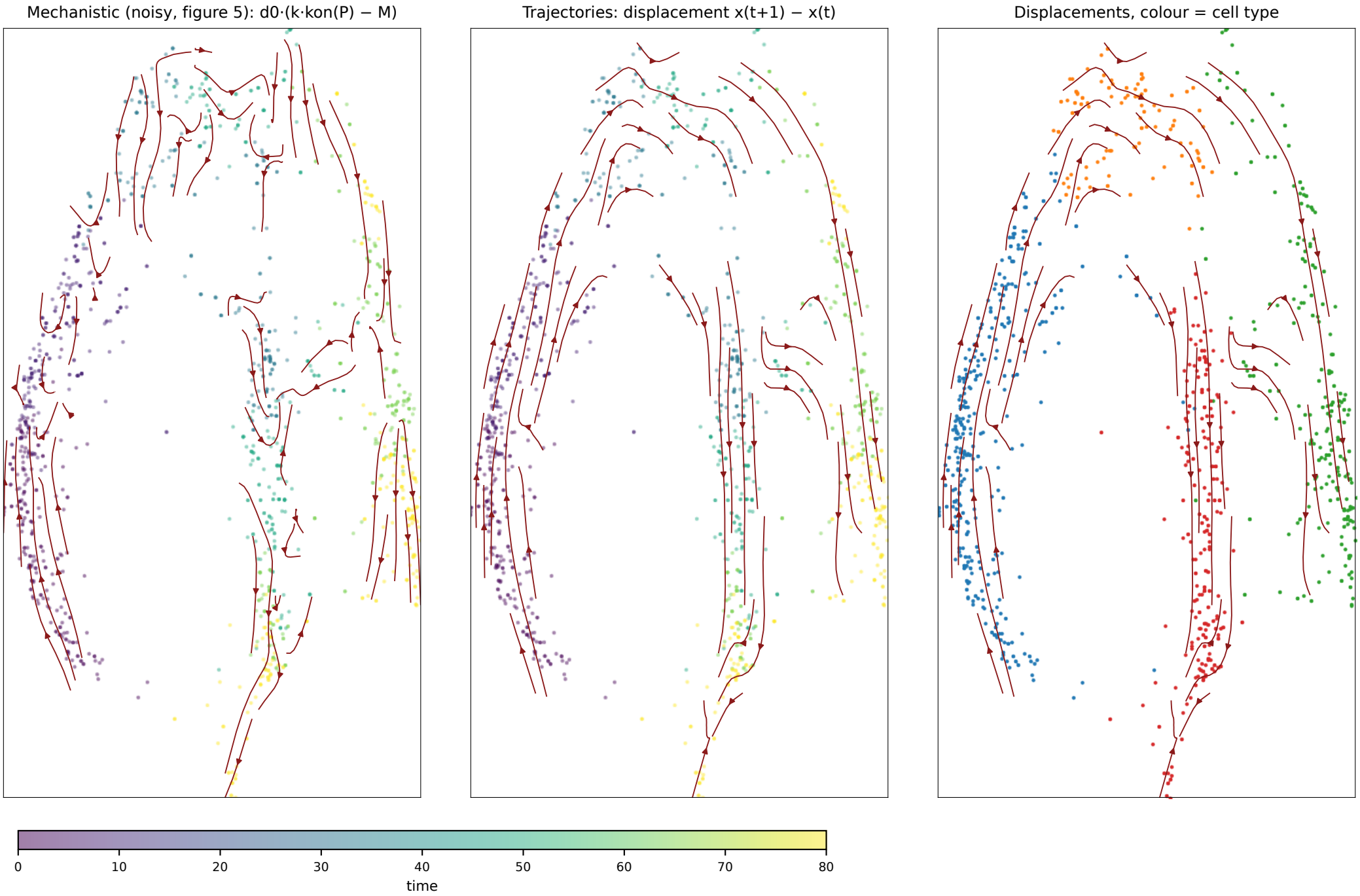
4. Protein dilution

Section unavailable:

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ValueError('protein_dilution = False, or no birth rate (obs proliferation_birth_rate / net rate)')
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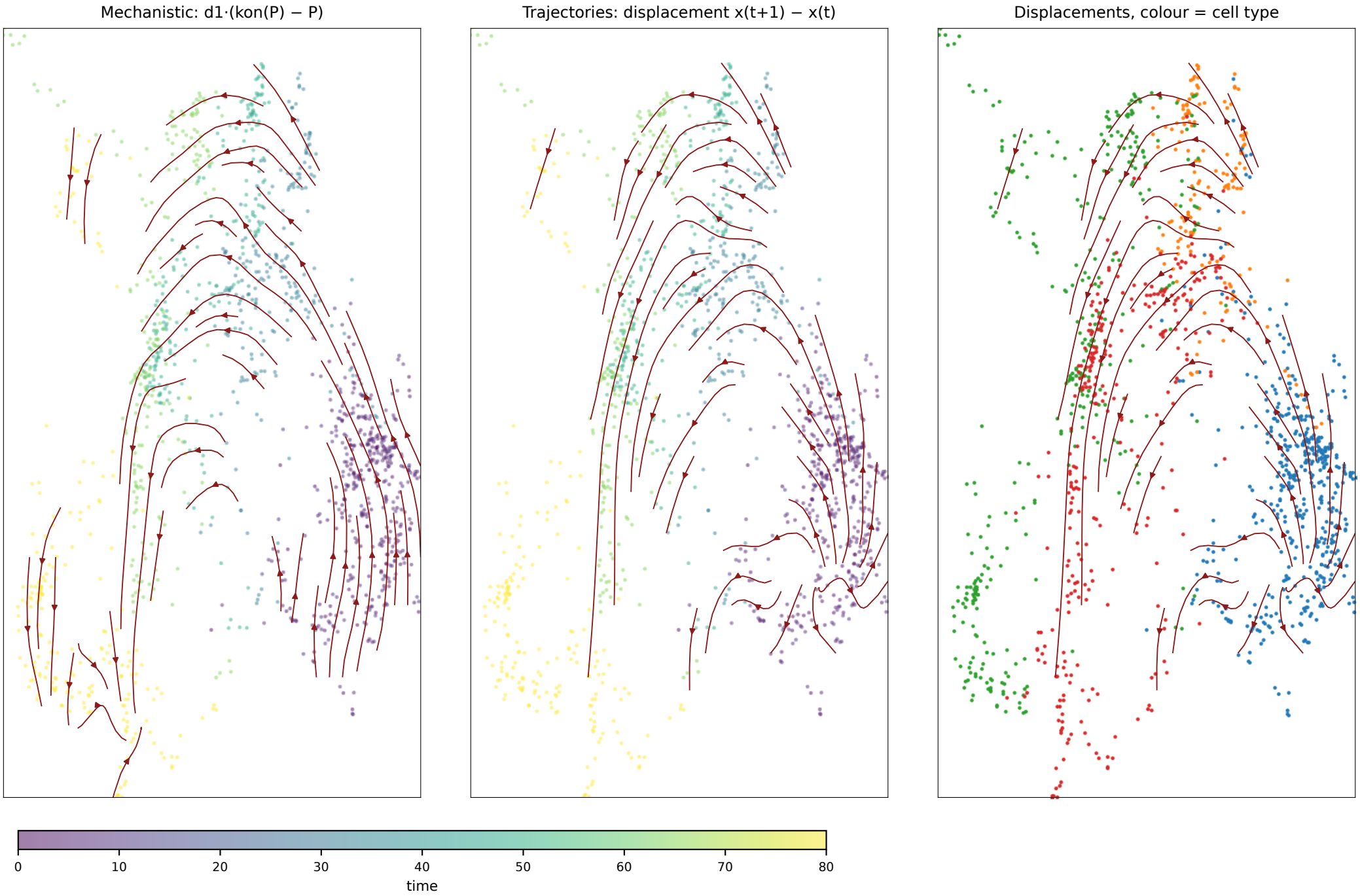
5. Learned dynamics — mRNA velocity and displacement fields

All 1260 trajectory states projected onto the PHATE learned on the observed mRNA (log1p, reference depth); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.24.



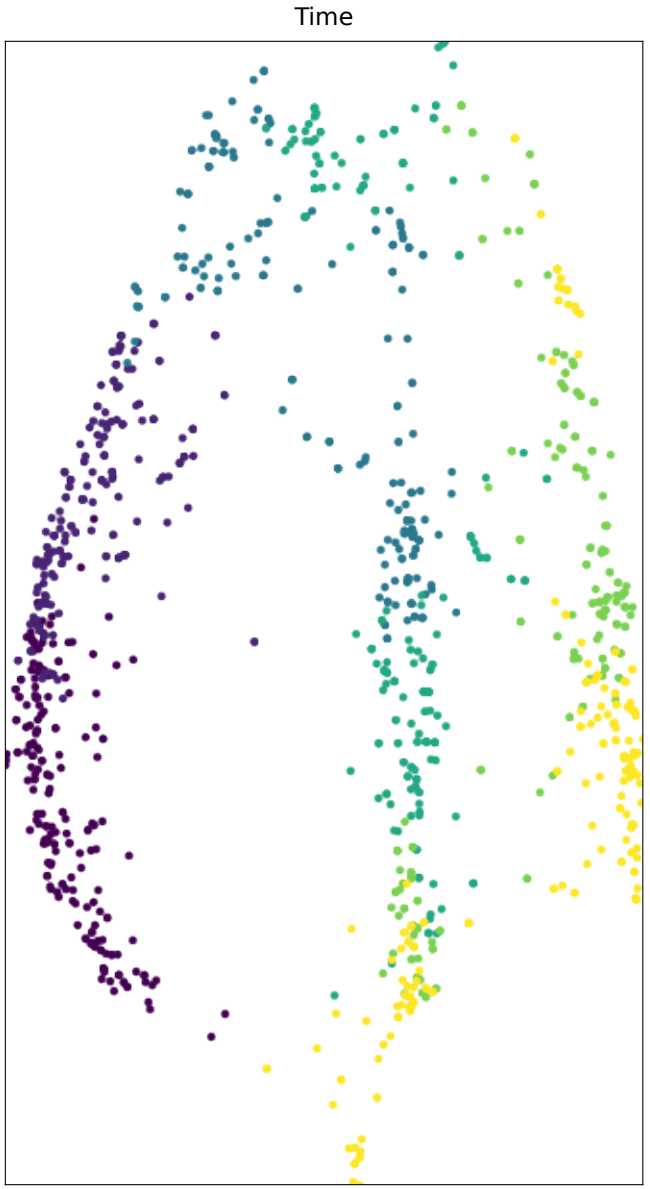
5. Learned dynamics — Proteins velocity and displacement fields

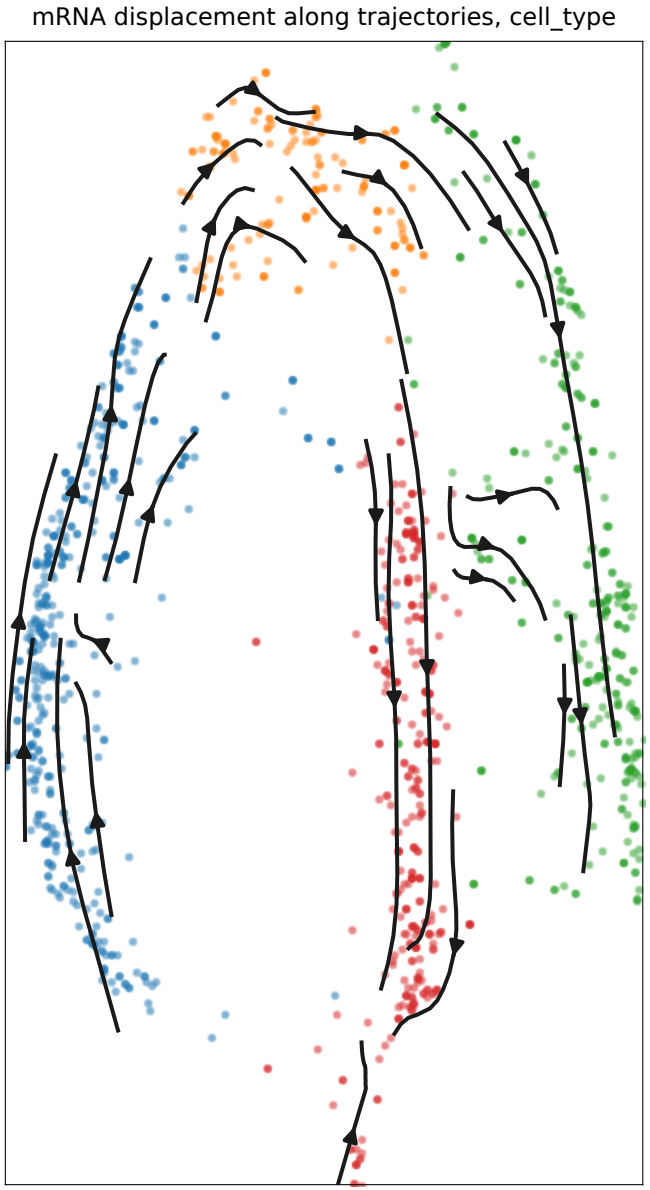
All 1260 trajectory states projected onto the PHATE learned on the protein trajectories (raw levels); kNN-transition projection of the fields. Mechanistic vs trajectory velocity agreement (weighted cosine, gene space): 0.58.



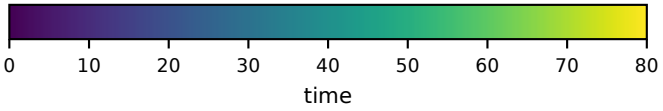
5. Learned dynamics — summary

All 1260 trajectory states on the PHATE learned on the observed mRNA (log1p, reference depth); displacement along the inferred trajectories x(t+1) − x(t); background of the field: cell_type.





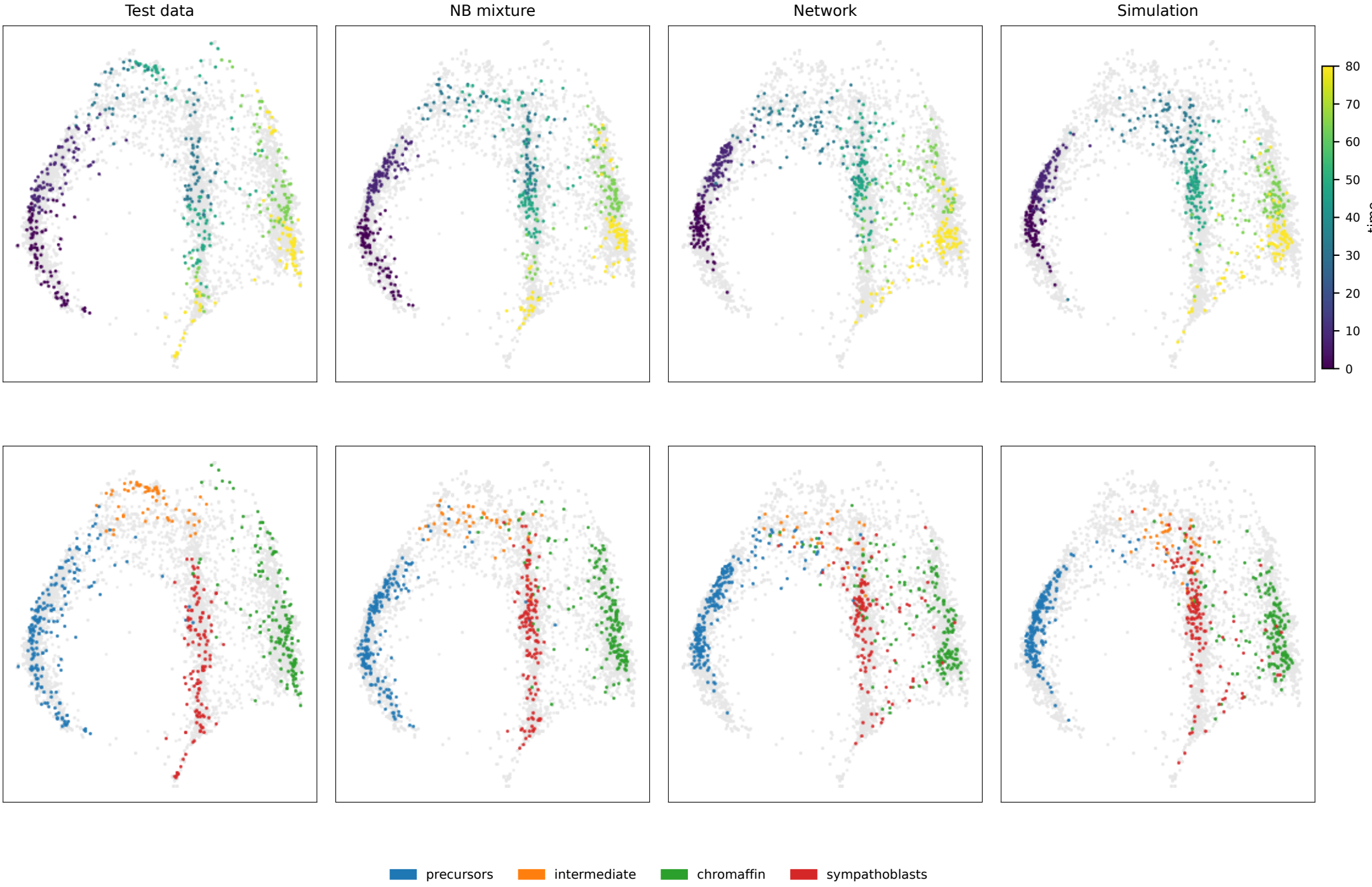
No net proliferation rate



precursors intermediate chromaffin sympathoblasts

6. Held-out test cells — predictions with the network fixed

Test cells classified into basins with the training mixtures, trajectories inferred with the training network fixed, then simulated; projected onto the PHATE learned on the observed data (grey: training).



6. Held-out test cells — quantitative checks

Per gene: 1-D Wasserstein distance between log1p counts, averaged over the common timepoints.

